Fiscal Year:	FY 2022	**Task Last Updated:**	FY 10/13/2021
PI Name:	Radlinska, Aleksandra Ph.D.		
Project Title:	Microgravity Investigation of Cement Solidification (MICS)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ara@engr.psu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	814-865-9427
Organization Name:	Pennsylvania State University		
PI Address 1:	Civil and Environmental Engineering		
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PI Web Page:			
City:	University Park	**State:**	PA
Zip Code:	16802-1408	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	11/22/2016	**End Date:**	11/21/2022
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Jackson, Enrique	**Contact Phone:**	
Contact Email:	enrique.m.jackson@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 11/21/2022 per NSSC information (Ed., 10/15/21) NOTE: End date changed to 11/21/2021 per NSSC information (Ed., 10/1/20)		
Key Personnel Changes/Previous PI:	October 2021 report: two undergraduate researchers were involved in research.		
COI Name (Institution):	Scheetz, Barry Ph.D. (Pennsylvania State University)		
Grant/Contract No.:	NNX17AC48G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The intent of this work is to utilize the microgravity environment aboard the International Space Station (ISS) to fully investigate the complex process of cement solidification. Microstructural development of hydrating cement occurs in stages during the hydration reaction and hardening process, which results in elaborate combinations of amorphous and crystalline phases. The morphology, volume fraction, and distribution of these phases ultimately determine the hardened cement's material properties. Minimizing gravity-driven phenomena, such as thermosolutal convective flow and sedimentation, will ensure crystal growth strictly by diffusion and a considerably different microstructure than that observed in typical laboratory conditions on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed microgravity experiments will satisfy two major objectives. First, the microstructure and material properties of the benchmark samples will be evaluated; different responses to thermal and mechanical loading are expected and will be characterized in detail. Any positive attributes found in the special environment will be reported with the intent of incorporating and improving Earth-based cement and concrete processing. Given the magnitude of cement usage worldwide, even a slight change in manufacturing or usage can have huge implications on the sustainability of the cement industry.
Task Progress:	Concurrently with the analysis of the ISS experiment samples, groundwork is being done on the development of an in-situ lunar concrete. A geopolymer concrete can be theoretically created almost entirely from in-situ lunar resources. The requirements for a geopolymer is an alkaline solution (such as sodium silicate and sodium hydroxide) and an aluminosilicate glassy source (the lunar regolith). Most of the solution can be theoretically produced from the regolith through ionic liquids and electrolysis methods. However, more research in that field is needed. The combination of the regolith and solution initiates a polymerization reaction to bind the regolith particles together. This is advantageous over traditional ordinary portland cement whose lunar derivation is compromised by both a dearth of calcium containing materials and extensive processing requirements. The development of the geopolymer lunar concrete includes both ground research and ISS investigations. The groundwork over the last year has focused on understanding how various lunar regolith simulants behave in such technology. No lunar regolith simulant is a perfect match to the actual lunar regolith so multiple trials are needed under the same experimental conditions to guide the design and implementation of such technology with actual lunar regolith. The ISS investigations include a durability study using the Materials International Space Station Experiment Flight Facility (MISSE-FF) and planning for a future solidification experiment on the ISS, similar to the past Microgravity Investigation of Cement Solidification (MICS) experimental setup.
Bibliography Type:	Description: (Last Updated: 10/02/2023)
Articles in Peer-reviewed Journals	Collins PJ, Grugel RN, Radlinska A. "Hydration of tricalcium aluminate and gypsum pastes on the International Space Station." Construction and Building Materials. 2021 May 24;285:122919. https://doi.org/10.1016/j.conbuildmat.2021.122919 , May-2021
Books/Book Chapters	Collins PJ, Grugel RN, Radlinska A. "The Influence of Variable Gravity on the Microstructural Development of Tricalcium Silicate Pastes." in "Earth and Space 2021: Materials, Structures, Dynamics, and Control in Extreme Environments. Proceedings of the 17th Biennial International Conference on Engineering, Science, Construction, and Operations in Challenging Environments, Virtual, April 19–23, 2021." Ed. P.J. van Susante, A.D. Roberts. American Society of Civil Engineers, 2021. p. 59-66. https://doi.org/10.1061/9780784483381.006 , Apr-2021
Significant Media Coverage	Strickland A. " 'Astronauts make concrete in space for the first time.' Article describing PI Aleksandra Radlinska's Microgravity Investigation of Cement Solidification (MICS) flight experiment." CNN site, September 11, 2019. https://edition.cnn.com/2019/09/10/world/astronaut-cement-space-station-scn-trnd/index.html , Sep-2019